How-To Library

Informatica

Creating an ODBC Data Source to Run a DB2 for z/OS Stored Procedure Transformation

© Copyright Informatica LLC 2010, 2021. Informatica

Abstract

This article describes how to create an ODBC data source that is required to run a Stored Procedure transformation for a DB2 for z/OS stored procedure in a PowerCenter mapping. It applies to PowerExchange 10.0 and later.

Supported Versions

- PowerExchange 10.0 or later

Table of Contents

Stored Procedure Transformations. . . . 2
Usage Considerations for Stored Procedure Transformations. . . . 2
Implementing a Stored Procedure Transformation in a Mapping. . . . 3
Creating an ODBC Data Source on Windows. . . . 3
Overview of the PowerExchange Data Source Wizard. . . . 4
PowerExchange Data Source Tab. . . . 4
General Tab. . . . 6
AS/400 Properties Tab. . . . 9
CAPX Properties Tab. . . . 10
CAPXRT Properties Tab. . . . 11
DB2/S390 Bulk Load Properties Tab. . . . 12

Stored Procedure Transformations

You can use Stored Procedure transformations for DB2 for z/OS stored procedures in a PowerCenter mapping. Use Stored Procedure transformations for read or write bulk data movement and change data capture (CDC) operations.

Usage Considerations for Stored Procedure Transformations

The following usage considerations apply to Stored Procedure transformations for DB2 for z/OS:

- You must create and configure an ODBC data source before you import the DB2 for z/OS stored procedure.
- PowerCenter imports and calls stored procedures with unqualified names. DB2 normally executes unqualified stored procedures according to the SQL CURRENT PATH value for PowerExchange. DB2 searches this list of schema names from left to right until it finds a schema name for which a stored procedure definition exists with the specified name.

  PowerExchange, however, executes any unqualified stored procedures by using the user ID of the PowerExchange connection as the schema name. So, if the user ID on the PowerExchange connection is USER1, PowerExchange calls unqualified stored procedures as USER1.*procedurename*.

  Note: To explicitly set the schema name for a Stored Procedure transformation, edit properties for the Stored Procedure transformation in the mapping.
- After you configure the Stored Procedure transformation in a mapping, use the PowerCenter Integration Service to run the session.

Implementing a Stored Procedure Transformation in a Mapping

To implement a Stored Procedure transformation for a DB2 for z/OS stored procedure in a mapping, complete the following high-level tasks:

1. Create and compile the DB2 stored procedure on the z/OS system.
2. Ensure that the PowerExchange Listener is running on the z/OS system.
3. Create and configure an ODBC data source on Windows. For more information, see the *PowerExchange Reference Manual*.
4. In the Transformation Developer, click **Transformation > Import Stored Procedure**.
5. In the **ODBC data source** box, enter or browse to the ODBC data source that you configured.
6. In the Mapping Designer, add the Stored Procedure transformation to the mapping.

   Note: When you import the stored procedure, the Mapping Designer does not qualify the stored procedure name with the owner. If you leave the name as is, PowerExchange executes any unqualified stored procedures by using the user ID of the PowerExchange connection as the schema name. So, if the user ID on the PowerExchange connection is USER1, PowerExchange calls unqualified stored procedures as USER1.*procedurename*.
7. If the schema of the stored procedure is different from the user ID that starts the PowerExchange Listener, edit properties for the Stored Procedure transformation in the mapping.

   On the **Properties** tab in the **Edit Transformations** dialog box, add the schema name to the stored procedure name in the **Stored Procedure Name** box. For example, if the stored procedure name is ZEMPS1 and the user ID that starts the PowerExchange Listener is USER2, enter the following value in the **Stored Procedure Name** box:

   ```
   USER2.ZEMPS
   ```

For more information about implementing Stored Procedure transformations in mappings, see the *PowerCenter Transformation Guide*.

Creating an ODBC Data Source on Windows

You can create an ODBC data source on Windows.

Note: You cannot install the PowerExchange ODBC drivers on Microsoft Vista.

To create an ODBC data source on Windows:

1. In **Control Panel**, double-click **Administrative Tools.**
2. In the **Administrative Tools** window, double-click **Data Sources (ODBC)**.
3. In the **ODBC Data Source Administrator** dialog box, click the **System DSN** tab.
4. On the **System DSN** tab, click **Add**.
5. In the **Create New Data Source** dialog box, select the **Informatica PowerExchange** driver from the list of available drivers and click **Finish**.

   If this driver is not available, see the *PowerExchange Installation Guide*.

   The **PowerExchange Data Source** wizard is displayed.
6. On the **PowerExchange Data Source** tab in the **PowerExchange Data Source** wizard, enter information to define the ODBC data source.

   Depending on the data source type that you select, enter values for all of the available properties.
7. For all data source types, you can optionally enter additional information on the **General** tab.

8. For the following data source types, enter information on the following tabs:

Data Source Type	Enter Information on
CAPX	CAPX Properties tab
CAPXRT	CAPXRT Properties tab
DB2 for i5/OS	AS/400 Properties tab
DB2 for z/OS	DB2/S390 Bulk Load Properties tab

9. Click **OK**.

   The ODBC data source appears in the **System Data Sources** list in the **ODBC Data Source Administrator** dialog box.

10. Click **OK**.

Overview of the PowerExchange Data Source Wizard

You can use the **PowerExchange Data Source** wizard to add a PowerExchange ODBC driver.

Based on the database type, you define different properties for the driver in the **PowerExchange Data Source** wizard. The following topics describe the tabs in the **PowerExchange Data Source** wizard.

PowerExchange Data Source Tab

On the **PowerExchange Data Source** tab in the **PowerExchange Data Source** wizard, enter the following properties to define a PowerExchange data source.

Name

A name for the data source.

Corresponding ODBC Parameter: Entry in ODBC.ini file

Location

The location of the data source.

The locations that appear in the **Location** list are defined by NODE statements in the dbmover.cfg configuration file on the local machine.

Corresponding ODBC Parameter: LOCATION

Type

The data source type.

Corresponding ODBC Parameter: DBTYPE

Local Codepage

Available code pages.

Default is **Default**, which causes the code page value in the PowerExchange configuration file or the default code page for the system to be used for the code page.

File Name

ADAUNLD data source type only. The name of the Adabas unload file.

Corresponding ODBC Parameter: DBQUAL1

Database

ADAUNLD, INFORMIX, MSSQL, ODBC or SYBASE data source type only.

The name of the database.

Corresponding ODBC Parameters: For an INFORMIX or SYBASE data source type, DBQUAL1.

For an ADAUNLD, MSSQL, or ODBC data source type, DBQUAL2.

DB Qual2

CAPX or CAPXRT data source type only.

Application name.

Corresponding ODBC Parameter: DBQUAL2

SSID or DBName

DB2, DB2UDB, or DB2400C data source type only.

The DB2 SSID or database name.

Corresponding ODBC Parameter: DBQUAL1

Write Mode

DB2, DB2UDB, DB2400C, MSSQL, NRDB, NRDB2, ODBC, or ORACLE data source type only.

The write mode.

The options are:

- **Confirm Write On.** Enables confirm write. Ensures that data sent to the PowerExchange ODBC driver is immediately sent to, rather than buffered by, the PowerExchange Listener. Additionally, a success or no success response is sent, which improves error recovery. However, sending a response can slow data transfer rates.
- **Confirm Write Off.** For DB2, DB2 UDB, and Oracle data sources only. Disables confirm write. Before using this setting, contact Informatica Global Customer Support.
- **Asynchronous With Fault Tolerance.** Enables asynchronous write with fault tolerance. For more information about asynchronous write, see the *PowerExchange Bulk Data Movement Guide*.

Default is **Confirm Write On**.

Corresponding ODBC Parameter: CONFIRMWRITE

Reject File

Available only if the **Asynchronous With Fault Tolerance** option is selected in the **Write Mode** list.

The location and file name of the reject file that is used during asynchronous write operations.

Enter a value of PWXDISABLE to disable reject file logging.

Maximum length is 384 characters.

For more information, see the *PowerExchange Bulk Data Movement Guide*.

Corresponding ODBC Parameter: REJECTFILE

Stop On Errors

Available only if the **Asynchronous With Fault Tolerance** option is selected in the **Write Mode** list.

The number of non-fatal errors that are allowed in the write phase before processing is terminated.

Valid values are 0 to 2147483647.

For more information, see the *PowerExchange Bulk Data Movement Guide*.

Corresponding ODBC Parameter: STOPONERRORS

SSID

DB2390IMG data source type only. The DB2 SSID.

Corresponding ODBC Parameter: DBQUAL1

Application Name

EMR data source type only. The application name used for the data extraction.

Corresponding ODBC Parameter: DBQUAL2

Wait Time

EMR, NRDB, or NRDB2 data source types only.

The maximum approximate time, in seconds, to wait for data before returning end-of-file (EOF).

The options are:

- 0. The EOF is returned when the end of the current log is reached. The end of the log is determined at the start of the extraction process, because the actual end point constantly changes.
- 2 to 86399. The number of seconds to wait for data before returning EOF.
- 86400. The EOF is never returned. The job waits forever.

Corresponding ODBC Parameter: WAITTIME

Server

INFORMIX or SYBASE data source type only.

The server on which the database is located.

Corresponding ODBC Parameter: DBQUAL2

DSN

MSSQL or ODBC data source type only.

The DSN name.

Corresponding ODBC Parameter: DBQUAL1

SQL *Net Name

ORACLE data source type only.

The SQL *Net name.

Corresponding ODBC Parameter: DBQUAL1

General Tab

On the **General** tab in the **PowerExchange Data Source** wizard, enter the following optional properties for a PowerExchange data source definition.

Integration Mask

The integration mask. If you specify DFN, the PowerExchange data source type is returned as follows:

```
DETAIL:NRDB
```

Default is blank, which returns the following string:

```
DETAIL
```

Informatica recommends using the default value.

Set this property only at the direction of Informatica Global Customer Support.

Corresponding ODBC Parameter: COMPATIBILITY

Maximum Rows

The maximum rows for retrieval.

Valid values are 0 to 2147483647.

Default is 0, which indicates unlimited rows.

Corresponding ODBC Parameter: MAXROWS

Pacing Size

The number of rows or kilobytes. Set this property when an application, such as an interactive application, cannot keep pace with the flow of data from the PowerExchange Listener.

For maximum performance, set this property to 0.

Default is the value of the API rows_requested parameter.

Corresponding ODBC Parameter: PACESIZE

As KB

Select this option to specify the pacing size in kilobytes.

Default is **As KB**.

Corresponding ODBC Parameter: INTERPRETASROWS

As Rows

Select this option to specify the pacing size in rows.

Default is **As KB**.

Corresponding ODBC Parameter: INTERPRETASROWS

Compress

Select this option to compress data.

This option is cleared by default.

Corresponding ODBC Parameter: COMPRESS

Encrypt

Select this option to encrypt data.

This option is cleared by default.

Corresponding ODBC Parameter: ENCRYPT, as follows:

- If this option is selected, ENCRYPT=N.

- If this option is cleared, ENCRYPT=Y.

Option

The encryption option, which is one of the following:

- **DES**. Data is encrypted by using Digital Encryption Standard (DES) algorithm.
- **RC2**. Data is encrypted by using the RSA security algorithm.

If you select an encryption option, you must select an encryption level in the **Level** list.

Corresponding ODBC Parameter: ENCRYPT, as follows:

- If the **DES** option is selected, ENCRYPT=DES.
- If the **RC2** option is selected, ENCRYPT=RC2.

Level

Required if you select the **DES** or **RC2** encryption option.

The encryption level.

The options are:

- **1**. Data is encrypted by using:
  - For DES: 56-bit key encryption
  - For RC2: 56-bit key encryption
- **2**. Data is encrypted by using:
  - For DES: Triple DES 164-bit key encryption
  - For RC2: 64-bit key encryption
- **3**. Data is encrypted by using:
  - For DES: Triple DES 164-bit key encryption
  - For RC2: 128-bit key encryption

Default is **1** if the **DES** or **RC2** encryption option is selected.

Corresponding ODBC Parameter: ENCRYPTLEVEL

MQ Get Opt

The type of read operation to be performed from an MQSeries queue.

The options are:

- **BROWSE**. Reads a copy of the data on the queue.
- **READ**. Reads the data and removes it from the queue.

Default is **READ**.

Corresponding ODBC Parameter: MQGETOPT

Application

The application name, which is used as part of the audit trail for data extractions by the PowerExchange Listener.

Corresponding ODBC Parameter: DTLAPP

AS/400 Properties Tab

For DB2 for i5/OS data sources only. On the **AS/400 Properties** tab in the **PowerExchange Data Source** wizard, enter the following properties to define a PowerExchange DB2 data source.

Isolation Level

The commit scope of the transaction.

The options are:

- **ALL**
- **CHG**
- **CS**
- **NONE**
- **RR**

This option is typically set to **CS**.

Default is **NONE**.

Corresponding ODBC Parameter: ISOLATION

Library List (Unqualified Table)

Space-delimited list of libraries that PowerExchange searches to qualify the first table name, if the table name is unqualified, on a select, insert, delete, or update statement. You can specify a maximum of 34 libraries. Each library name is a maximum length of ten characters.

Note: If you specify a library list in both the **Library List (Unqualified Table)** and **Table/File Override** boxes, and a table is found in both lists, the table found in the **Table/File Override** list takes precedence.

Corresponding ODBC Parameter: LIBRARYLIST

Table/File Override

Space-delimited list of file overrides in the following format:

```
filename/newlibrary/newfilename/newmembername
```

Alternatively, you can use the following format:

```
filename/newlibrary/newfilename/
```

In this format, the member defaults to *FIRST.

You can specify a maximum of eight files. Each file specification is a maximum length of 43 characters, as follows:

- Ten characters for the *filename* value
- Ten characters for the *newlibrary* value
- Ten characters for the *newfilename* value
- Ten characters for the *newmember* value, if specified
- Three characters for the slash (/) separators

When you use this override, any occurrence of the specified file name in a SQL statement is overridden with the *newlibrary*/*newfilename*/*newmembername* combination, regardless of whether the file is qualified or not. This includes any files that are specified in the **Library List (Unqualified Table)** override.

Note: If you specify a library list in both the **Library List (Unqualified Table)** and **Table/File Override** boxes, and a table is found in both lists, the table found in the **Table/File Override** list takes precedence.

Corresponding ODBC Parameter: OVRDBF

CAPX Properties Tab

On the **CAPX Properties** tab in the **PowerExchange Data Source** wizard, enter the following properties to define a PowerExchange CAPX data source.

Extract Type

The extraction type. The options are:

- **SL**. Start the extraction from the point where the last extraction ended, either by restart tokens for z/OS, or by checkpoint timestamps for i5/OS.
- **RS**. Restart the extraction from the start point of the last extraction. This option enables you to run the same extraction multiple times.

Corresponding ODBC Parameter: CAPXTYPE

Image Type

The capture image type. The options are:

- **BA**. Captures the before image data, before the latest change, and the after image data, after the latest change. The before image data is shipped with an action character forcing the deletion of the record. The after image data, which includes the change data, is in the format of an insert operation.
- **AI**. Captures the after image data only.
- **TU**. Captures the before and after image data, as it occurs. The before image data enables the application to ensure that no changes occurred in the record since the request. The after image data appears as an update operation. The update is a full image, but the application that captures this change can change the target record rather than delete it and insert a new record, as occurs with a BA.

Corresponding ODBC Parameter: CAPXIMAGETYPE

AS400 Library/Filename

Overrides the fully qualified DB2 for i5/OS library and table name specified in the PowerExchange capture registration.

For example:

```
STQA/NEWTABLE
```

Corresponding ODBC Parameter: DBQUAL1

AS400 Instance

Overrides the name of the DB2 for i5/OS instance specified in the PowerExchange capture registration.

Corresponding ODBC Parameter: DBQUAL2

CAPI Connection

Overrides the default database connection specified in CAPI_CONNECTION statement in the DBMOVER configuration file.

To point to the relevant data source, specify the value in the NAME parameter of the CAPI_CONNECTION statement in the DBMOVER configuration file.

Corresponding ODBC Parameter: DTLCONN_OVR

Extraction Schema

Overrides the schema name specified in an extraction map.

Maximum length is 128 characters. No spaces are allowed.

Corresponding ODBC Parameter: XTRASCHEMA

CAPXRT Properties Tab

On the **CAPXRT Properties** tab in the **PowerExchange Data Source** wizard, enter the following properties to define a PowerExchange CAPXRT data source.

Extract Type

The extraction type. The options are:

- **SL**. Extracts all data since the last extraction.
- **RS**. Restarts the previous, or a specified, extraction.

Corresponding ODBC Parameter: CAPXTYPE

Image Type

The image type. The options are:

- **BA**. Captures the before image data, before the latest change, and the after image data, after the latest change. The before image data is shipped with an action character forcing the deletion of the record. The after image data, which includes the change data, is in the format of an insert operation.
- **AI**. Captures the after image data only.
- **TU**. Captures the before and after image data, as it occurs. The before image data enables the application to ensure that no changes occurred in the record since the request. The after image data appears as an update operation. The update is a full image, but the application that captures this change can change the target record rather than delete it and insert a new record, as occurs with a BA.

Corresponding ODBC Parameter: CAPXIMAGETYPE

AS400 Library/Filename

Overrides the fully qualified DB2 for i5/OS library and table name specified in the PowerExchange capture registration.

For example:

```
STQA/NEWTABLE
```

Corresponding ODBC Parameter: DBQUAL1

AS400 Library/Journal

Overrides the fully qualified library and journal name specified in the PowerExchange capture registration.

For example:

```
STQA/NEWJOURNAL
```

Corresponding ODBC Parameter: UAP

Oracle Instance

Overrides the Oracle instance value in the second positional parameter in the ORACLEID statement in the dbmover.cfg configuration file. For example:

```
ORACLEID=(coll_id,oracle_sid,connect_string,cap_connect_string)
```

Used in conjunction with the **Oracle Connection String** value to enable the use of a single set of capture registrations to capture data from multiple Oracle instances.

If you specify an **Oracle Instance** value, but no **Oracle Connection String** value, Oracle capture uses the Oracle connection string value specified in the ORACLEID statement in the dbmover.cfg file.

Corresponding ODBC Parameter: ORAINST

Oracle Connection String

Overrides the Oracle connection string value in the third positional parameter in the ORACLEID statement in the dbmover.cfg configuration file. For example:

```
ORACLEID=(coll_id,oracle_sid,connect_string,cap_connect_string)
```

Used in conjunction with the **Oracle Instance** value to enable the use of a single set of capture registrations to capture data from multiple Oracle instances.

If you specify an **Oracle Connection String** value, but no **Oracle Instance** value, Oracle capture uses the Oracle connection string value specified in the ORACLEID statement in the dbmover.cfg file.

Corresponding ODBC Parameter: ORACONN

Oracle Schema

Overrides the Oracle schema value in a PowerExchange registration group to enable the use of a single set of capture registrations to capture data from multiple schemas that might exist in an Oracle instance.

Corresponding ODBC Parameter: ORASCHEMA

IDMS LOGSID

For IDMS synchronous only. Overrides the instance value specified in the LOGSID statement in the dbmover.cfg configuration file, and the instance value specified in the PowerExchange capture registration.

Corresponding ODBC Parameter: LOGSID

DB2 UDB Database

For DB2 for Linux, UNIX, and Windows only. Required for EMR only. Overrides the database connection value specified in the PowerExchange extraction map.

Corresponding ODBC Parameter: DATABASE

CAPI Connection

Overrides the default database connection specified in a CAPI_CONNECTION statement in the DBMOVER configuration file.

To point to the relevant data source, use the value that is specified in the NAME parameter of the CAPI_CONNECTION statement in the DBMOVER configuration file.

Corresponding ODBC Parameter: DTLCONN_OVR

Extraction Schema

Overrides the schema name specified in a PowerExchange extraction map.

Maximum length is 128 characters. No spaces allowed.

Corresponding ODBC Parameter: XTRASCHEMA

DB2/S390 Bulk Load Properties Tab

For DB2 data sources only. On the **DB2/S390 Bulk Load Properties** tab in the **PowerExchange Data Source** dialog box, enter the following properties to define PowerExchange for DB2 bulk load processing.

Bulk Load

Select this option to enable input on this page.

Corresponding ODBC Parameter: BULKLOAD

Load Options

The load option.

The options are:

- **INSERT**. Insert.
- **REPLACE**. Replace.

Corresponding ODBC Parameter: LOADOPTIONS, as follows:

- INSERT is equivalent to LOADOPTIONS=1.
- REPLACE is equivalent to LOADOPTIONS=4.

Delete Temporary Files

Indicates whether to delete temporary files. The options are:

- **NO.** Temporary files are not deleted.
- **BEFORE**. Temporary files are deleted before running the loader.
- **AFTER SUCCESS ONLY**. For a mode type of **JOB** only. Temporary files are deleted after running the loader, if the return code is 0.
- **AFTER**. For a mode type of **JOB** only. Temporary files are deleted after running the loader.

Corresponding ODBC Parameter: DELETETEMPFILES, as follows:

- NO is equivalent to DELETETEMPFILES=1.
- BEFORE is equivalent to DELETETEMPFILES=2.
- AFTER SUCCESS ONLY is equivalent to DELETETEMPFILES=3.
- AFTER is equivalent to DELETETEMPFILES=4.

JCL Template

The name of the JCL file template on the target system, which is a member in a PDS defined on the system where the PowerExchange Listener is running.

Default is DB2LDJCL. You must customize this JCL before using it.

Corresponding ODBC Parameter: JCLTEMPLATE

CTL Template

The name of the DB2 bulk load control file template on the target system, which is a member in a PDS defined on the system where the PowerExchange Listener is running.

If you specify a CTL template, the **Load Options** option is ignored.

Default is DB2LDJCL. You must customize this JCL before using it.

Corresponding ODBC Parameter: CTLTEMPLATE

Mode Type

The mode type. The options are:

- **JOB**. Submits as a job to the internal reader.

- **NOSUBMIT**. Not submitted. Instead, creates the DB2 loader-specific control files and data files, saves them on the target system, and generates an executable JCL deck on disk.
- **TASK**. Submits as a subtask of the PowerExchange Listener.

Corresponding ODBC Parameter: MODETYPE as follows:

- TASK is equivalent to MODETYPE=1.
- NOSUBMIT is equivalent to MODETYPE=2.
- JOB is equivalent to MODETYPE=3.

Mode Time

The mode time. The options are:

- **WAIT**. Synchronous. For a mode type of **JOB** or **TASK** only. Report the results at end the end of the job or task.
- **NOWAIT**. Asynchronous. For a mode type of **JOB** or **NOSUBMIT** only. Returns OK when a job is submitted.
- **TIMED**. Synchronous. For a mode type of **JOB** only. Wait for the specified number of seconds.
- **DATAONLY**. For all mode types, but typically used with the **NOSUBMIT** mode type. Creates the data DB2 loader file.

Corresponding ODBC Parameter: MODETIME, as follows:

- WAIT is equivalent to MODETIME=1.
- NOWAIT is equivalent to MODETIME=2.
- TIMED is equivalent to MODETIME=3.
- DATAONLY is equivalent to MODETIME=4.

Time

Valid only for a mode type of **JOB** and a mode time of **TIMED**.

A value from 0 to 99999, as follows:

- **0**. No wait.
- **99999**. Wait.
- Any other value. The number of seconds to wait.

Corresponding ODBC Parameter: TIME

Space

The z/OS space allocation. The options are:

- **TRACK**. Tracks.
- **CYLINDERS**. Cylinders.

Corresponding ODBC Parameter: SPACE

Primary Space

Amount of primary space for z/OS files.

Corresponding ODBC Parameter: PRISPACE

Secondary Space

Amount of secondary space for z/OS files.

Corresponding ODBC Parameter: SECSPACE

FName

The file name on the host system. Used as the stem for files created on z/OS.

The files created are:

- FNAME.DTL*nnnnn*

  Data for load.
- FNAME.DTL*nnnnn*.CTL

  Load commands.
- FNAME.DTL*nnnnn*.SQL

  SQL template - create table.
- FNAME.DTL*nnnnn*.SYSPRINT

  Output from loader.

Corresponding ODBC Parameter: FNAME

Author

Diane Fleming